

Serial No: 0531401 Sub Code: 23UCS1 / 23UCA1
H.H.THE RAJAH'S COLLEGE (AUTONOMOUS B), PUDUKKOTTAI -622 001
B.C.A. DEGREE EXAMINATIONS – NOVEMBER - 2024
PART III - COMPUTER SCIENCE / B.C.A. - CORE COURSE-I
PYTHON PROGRAMMING

Time: 3 hours

Max: 75 marks
(10x2=20marks)

Section – A

Answer ALL the Questions

1. What is the feature and applications of Python?
2. Define Python Literal.
3. Write the syntax and example for While Loop in Python.
4. Compare and contrast if-else and nested-if with example.
5. Define function in Python with syntax.
6. What is Recursion in Python?
7. Define List.
8. What is meant by Tuples in Python?
9. What are the different types of files in Python?
10. How to do two splits in Python?

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Section – B

(5x5=25marks)

Answer ALL the Questions

11. a) Write a Python program to convert Seconds into Hours, Minutes and Seconds.

(OR)

- b) Describe in detail about Python Built-in data types with example.

12. a) Solve Sales Commission of representative sales :

- i) if sales ≤ 500 then no commission
- ii) if sales > 500 and sales ≤ 5000 then commission = 10% of sales
- iii) if sales > 5000 then commission = 500 + 8% of sales

Write a Python program using else if ladder.

(OR)

(1)

b) Write a Python program to print in a Pyramid Format using For loop

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

13. a) Write a Python program to find Simple Interest using function.

(OR)

b) Illustrate in brief about Python dir() function syntax with example.

14. a) Explain in detail about the following List operations

- | | |
|------------------------|--------------------------|
| i) Access List Items | ii) Range of Indexes |
| iii) Change Item Value | iv) Loop Through a List. |

(OR)

b) Examine in detail about the following Tuple operations.

- | | |
|--------------------------|---------------------------|
| i) Create Tuple | ii) Access Tuple Items |
| iii) Change Tuple Values | iv) Loop Through a Tuple. |

15. a) Illustrate in detail about opening and closing files in Python syntax with example.

(OR)

b) Difference between write() and writelines() function in Python syntax with example.

Section – C

(3x10=30marks)

Answer any THREE Questions Only

16. Explain in brief about various operators used in Python with example.

17. Examine in detail about various Jump statements in Python syntax with example.

18. Discuss the following String function syntax with example.

- i) islower() method
- ii) isupper() method
- iii) join() method
- iv) maketrans() method
- v) title() method.

19. Analyze in brief about the following Dictionaries Operations

- i) Create and print a Dictionary
- ii) Access items in a Dictionary
- iii) Loop through a Dictionary
- iv) Dictionary Length
- v) Removing items in a Dictionary using pop() method.

20. Write a Python program to read first n lines of a file.

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(3)